

Full Test — Answers

Final answers with a one-line reason. For full methods, revisit the concept notes.

Objective

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- Q1** (c) **8.** Three gaps, each + or -: 2^3 .
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- Q2** (c) **4n.** $4n = 2(2n)$, always even.
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- Q3** (b) **the sum of its digits is divisible by 9.**
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- Q4** (b) **14.** 7×29 to 7×42 .
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- Q5** **1.** $7+3+0+9 = 19 \rightarrow 10 \rightarrow 1$.
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- Q6** **True.** One of two consecutive numbers is even.
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- Q7** (c) **9.** Digit sum $4+0+5 = 9$.
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- Q8** (d) **A false, R true.** 1 has only one factor, so it is not prime.
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Short answer

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- Q9** **k-1, k-2, k-3, k-4, k-5, k-6.**
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- Q10** **Smallest 1008, greatest 9999.**
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- Q11** **z = 0 or 9.** $9 + z$ must be a multiple of 9.
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- Q12** **Always even.** $2a + 2b = 2(a + b)$, a multiple of 2.
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Long answer

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- Q13** **7, 8, 9, 10.** $4n + 6 = 34 \Rightarrow n = 7$.
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- Q14** **Divisible by 3 only.** Digit sum 30 ($\div 3$ yes, $\div 9$ no); ends in 9 (not 2); alternating sum 4 (not 11).
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- Q15** **A = 3, B = 7, C = 1.** $37 \times 3 = 111 = 3 \times 37$.
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- Q16** **Digital root = 2.** $63p$ and $18q$ are multiples of 9; 29 leaves remainder 2, so the root is 2 for all p, q .
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Total: 28 marks.